RANKS OF TENSOR PRODUCTS AND OF HOM MODULES OVER A DOMAIN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tensoring with the fraction field commutes with Hom out of a finitely presented module. Algebra, Lemma \ref{algebra-lemma-hom-from-finitely-presented}, cited at proof lines 5064-5066. Identifies the tensor product of the Hom module with the Hom module of the tensor products.

Source reference. Original lemma 0H9X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21542–21542. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0583: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6766–6766. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a domain. Let M and N be R-modules.*

(1) *The rank of $M \otimes_R N$ is the product of the ranks of M and N.*
(2) *If M is a finitely presented R-module, then the rank of $\mathrm{Hom}_R(M, N)$ is the product of the ranks of M and N.*

Proof. Part (1) follows from the same fact for vector spaces. Assume M is finitely presented as an R-module. Then $\mathrm{Hom}_R(M, N) \otimes_R K$ is isomorphic to $\mathrm{Hom}_K(M \otimes_R K, N \otimes_R K)$ as a K-vector space by Algebra, Lemma 0583. Since also $M \otimes_R K$ is finite dimensional, we conclude from the corresponding fact for vector spaces. □

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